GridLedger

Technical & Forensic Report

ASSET	Nabiwi Chitsazo Maize Mill, M1 Corridor, Malawi
PERIOD	32 months | July 2023 – March 2026
DATE	April 2026
REF	T.G. Msonda & Associates – Technical & Forensic Audit
STATUS	Institutional Grade | Audit-Hardened | Final

LEAD STATEMENT	*GridLedger classifies energy-backed production and associated cashflows into VERIFIED, LEAKAGE, and UNVERIFIED states using independently auditable energy, production, and payment data. Only VERIFIED cashflows are considered financeable.*

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1. Purpose

This report documents the verification methodology, data sources, and forensic findings of the GridLedger system applied to the Nabiwi maize mill. The objective is to provide an independent, audit-ready record of the mill's energy consumption, production efficiency, and cash realisation – establishing a financeable asset profile.

NOTE

This report is prepared as a supporting technical document and does not constitute a formal property valuation under Malawi valuation standards. GridLedger does not verify physical maize throughput independently and relies on energy-to-output modelling validated through historical consistency.

2. System Overview

GridLedger is an enforcement protocol that applies deterministic energy constraints with probabilistic operator reporting, reconciled through independent payment rails. It does not rely on operator-reported data alone. Instead, it cross-validates three independent data streams:

Data Stream	Source	Integrity Classification
Energy consumption	ESCOM prepaid token receipts (meter 37154463253)	Utility-originated. Treated as primary high-integrity source.
Production reports	Operator SMS logs (structured, timestamped)	Non-authoritative. Used only when aligned with independent energy and payment records.
Cash remittance	Airtel Money receipts (transaction IDs, agent names)	Independent payment rail. Corroborating anchor for revenue verification.

Control Hierarchy – Why This Is Enforceable

Step	State	Implication
1	ESCOM token issued	Production permission granted
2	No token	No grinding possible
3	No grinding	No revenue generated
4	Token control = revenue control	GridLedger sits at the point of energy release

ENFORCEMENT

GridLedger does not enforce behaviour at the point of reporting. It enforces behaviour at the point of energy release, where production dependency creates compliance. Outputs represent constrained economic estimates under controlled energy conditions, not direct physical measurement of throughput.

Variance outside established efficiency bands triggers classification downgrade and enforcement intervention.

3. Data Extracted

From 27,631 SMS messages (HiSuite export, 14 March 2026) and 481 ESCOM token purchase records, the following metrics were derived:

Metric	Value
Total energy consumed	31,329 kWh
Total reported revenue (SMS)	MWK 25,115,000
Verified node efficiency (field-tested)	22.66 kg/kWh
Metered energy not reconciled to reports	6,833 kWh
Corresponding unobserved production	155 metric tonnes (model-derived)
Model-derived production variance	MWK 9.3M – 13.6M equivalent (non-cash, non-claimable)

MODEL NOTE *The MWK 9.3M – 13.6M figure is an analytical reconstruction based on verified efficiency ratios and should not be interpreted as realised or collectible revenue. Under GridLedger enforcement, such discrepancies are systematically eliminated through pre-allocation reconciliation requirements.*

Classification Categories

Category	Description
VERIFIED	Energy, production, and Airtel cash match within 2% tolerance. Financeable.
LEAKAGE	Documented operational deductions — traceable and controllable by owner.
UNVERIFIED	No Airtel receipt within 48h. Data gap — not proven loss.
UNOBSERVED (Model-Derived)	Analytical estimate based on metered energy variance. Not cash-verified.

4. March 2026 Forensic Summary

A detailed analysis of March 2026 (19 production days) provides the most current performance snapshot and serves as the primary basis for financeability assessment.

Category	Days	Revenue (MWK)	% Total	Note
VERIFIED	9	1,029,600	64%	Fully reconciled. Bankable.
LEAKAGE	3	40,000	2.5%	Traceable deductions. Controllable.
UNVERIFIED	7+	587,700	—	No digital receipt within 48h.

Summary Metric	Value
Total Reported Revenue (March)	MWK 1,617,300
VERIFIED + LEAKAGE	MWK 1,069,600 (66%)
Financeable (VERIFIED only)	MWK 1,029,600 (64%)

ENFORCEMENT RULE

Under GridLedger enforcement rules, any UNVERIFIED cycle blocks subsequent token allocation until resolved. LEAKAGE cycles are visible and controllable — the owner can authorise or prohibit such deductions through the system. Only VERIFIED cashflows are included in financeability calculations.

Operator compliance risk is mitigated through enforced energy gating and real-time cash verification.

5. Five-Cycle Clean Verification Table

These cycles are presented as proof-of-capability, not as a representation of average performance (see Section 4 for full March 2026 distribution).

Date	Token (kWh)	Rpt (kWh)	Exp Rev (MWK)	SMS (MWK)	Airtel (MWK)	CRR	Airtel Trans ID	Agent
2026-03-02	59.9	60	81,000	81,000	81,000	1.00	CI260302.1017.C90092	ESTERE PHIRI
2026-03-03	59.9	60	81,000	81,000	81,000	1.00	CI260303.0914.Y06496	CHARLSE KWENDA
2026-03-06	59.9	60	81,000	81,000	81,000	1.00	CI260306.0739.B43633	ESTERE PHIRI
2026-03-07	59.9	60	81,000	81,000	81,000	1.00	CI260307.1626.L16276	VICTOR MUMBA
2026-03-10	59.9	60	81,000	81,000	81,000	1.00	CI260310.1124.D31118	DAVIE MARTIN

VERIFICATION BASIS

Each cycle above is backed by an ESCOM token receipt and an Airtel Money confirmation. The complete list of token IDs and transaction IDs is available in Appendix A. Cash Realisation Ratio (CRR) = Airtel Cash / Expected Revenue.

6. Cycle Proof Sheets — Closed-Loop Evidence

The following two cycles are presented with full chained evidence to demonstrate the closed-loop verification architecture: ESCOM token → SMS production report → Airtel Money receipt.

Cycle — 2026-03-02

Element	Detail
ESCOM Token ID	`59407951658817740201`
Token Size	59.9 kWh
SMS Evidence	`Pa 2.3.26 Open 60 Close 0 Ndagayil 60 Units Amount 81,000 Late 1350`
Airtel Trans ID	`CI260302.1017.C90092`
Airtel Amount	MWK 81,000
Agent	ESTERE PHIRI
Result	**Energy → Production → Cash = CLOSED LOOP VERIFIED**

Cycle — 2026-03-07

Element	Detail
ESCOM Token ID	59407951658817740247
Token Size	59.9 kWh
SMS Evidence	Pa 7.3.26 Open 60 Close 0 Ndagayil 60 Units Amount 81,000 Late 1350
Airtel Trans ID	CI260307.1626.L16276
Airtel Amount	MWK 81,000
Agent	VICTOR MUMBA
Result	**Energy → Production → Cash = CLOSED LOOP VERIFIED**

7. Indicative Financeability

Based on the VERIFIED cashflow of March 2026 (MWK 1,029,600/month), the asset supports a conservative lending estimate. These figures are derived solely from independently verified cashflows, not from projections or operator assertions.

Metric	Value	Basis
Annualised VERIFIED cashflow	**MWK 12.36M**	Based on March 2026 VERIFIED cycles extrapolated over 12 months.
Working capital facility (2.0x–2.5x coverage)	**MWK 5M – 6M**	Conservative range at current verified performance.
Upside under full enforcement (85–95% verification)	**MWK 8M – 10M**	Achievable if UNVERIFIED cycles are resolved through gating.

CONDITIONS

All financeability assumptions are conditional upon uninterrupted GridLedger enforcement and do not represent historical lending eligibility under uncontrolled operating conditions. All financeability assumptions are contingent upon exclusive or delegated control over ESCOM token provisioning. Without this control, GridLedger reverts to an observational system and cannot guarantee cashflow enforcement.

The VERIFIED cashflow profile may support consideration for working capital facilities, subject to independent lender assessment and credit policy.

8. System Guarantees & Failure Containment

System Guarantees

Guarantee	Detail
Idempotent token allocation	No double allocation on retry
Time-weighted risk	Delays penalised hourly via step-function multiplier
Effective rate tracking	Logs effective_rate_per_kwh per cycle

Real-time classification	VERIFIED / LEAKAGE / UNVERIFIED computed automatically
Energy allocation control	Governed by predefined enforcement rules; supports delegated or multi-party authorisation in institutional deployments

Failure Containment Mechanism

Trigger	Enforcement Response
Operator non-compliance	Immediate token suspension
Payment rail disruption	Revenue freeze at source (energy denial)
Data reconciliation failure	Manual override required by asset owner

LOSS CONTAINMENT

Maximum loss exposure per non-compliant event is bounded to a single production cycle (≤59.9 kWh equivalent), after which energy access is programmatically suspended. This ensures loss exposure is time-bounded and does not compound across cycles. No new capital is deployed into the system without prior cycle verification.

The system does not assume revenue. It classifies revenue based on independent evidence.

9. System Limitations

GridLedger operates within the following constraints. These limitations are accepted as boundary conditions; the system is designed to flag and escalate anomalies, not to eliminate all possible evasion.

Limitation	Description
ESCOM supply dependency	The system cannot enforce energy allocation during prolonged grid outages.
Airtel Money availability	Cash verification requires mobile money transaction confirmations.
No direct output measurement	Production is modelled from energy consumption, not weighed in real time.
Off-grid parallel production	Diesel generators or secondary meters could bypass energy control.
Operator collusion	The system does not prevent side agreements or external transactions that bypass the formal cash reporting channel.

10. Independent Verification Pathway

All primary data referenced in this report can be independently verified through the following primary sources. GridLedger does not originate data; it indexes and reconciles independently generated records.

Source	Reference	Role
ESCOM prepaid token records	Meter #37154463253	Energy purchase authority. Utility-grade, tamper-evident.
Airtel Money transaction records	Transaction IDs + agent logs	Independent payment rail. Bank-grade confirmation.
Raw SMS export file	HiSuite dataset, timestamped	27,631 messages. Operator production reports.

11. Conclusion

The GridLedger protocol has successfully transformed the Nabiwi maize mill from an opaque, cash-based operation into a transparent, verifiable, and financeable asset over a 32-month period. The system has:

Achievement	Detail
Energy-to-output ratio established	22.66 kg/kWh (field-verified)
Historical data reconstructed	32 months of production and cash data (651 production days)
Enforcement architecture live	VERIFIED / LEAKAGE / UNVERIFIED classification operational
Glass Box certification path	Pending ten consecutive clean production cycles
Capital signal created	Portable credit signal for financial institutions upon certification

STRATEGIC IMPLICATION	*GridLedger converts informal, cash-based production nodes into verifiable economic units capable of supporting structured finance. This enables capital deployment into previously unfinanceable assets by replacing trust with enforcement. The asset is no longer dependent on operator credibility; it is governed by verifiable energy-to-cash conversion constraints.* GridLedger does not track money. It controls the conditions under which money can exist.

Appendices

Appendix	Contents
Appendix A	Full cycle proof sheets (selected cycles) + full token ID list
Appendix B	Full list of Airtel Money transaction IDs (March 2026)
Appendix C	Monthly production summaries (July 2023 – March 2026)
Appendix D	Reconciliation CSV (651 production days)

Appendices available on request from GridLedger Enforcement Protocol.